SYSTEM ARCHITECTURE · SCIENTIFIC DATA · APPLIED AI

Hanno Knatz

I design and build software and data systems for scientific work. At BioNTech, I led a Lab IT engineering team and owned the architecture and lifecycle of a biological assay analysis platform used across global laboratories.

Today, I develop independent projects in AI agents, local inference and workflow automation.

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SELECTED PROJECTS

Independent work since Apr 2026.

Bullshitbox

EXPERIMENT

Computer-use agents

Bullshitbox is an experimental platform for computer-use agents — LLM agents that see a screen and operate software. A testbed for building and evaluating agentic workflows with tool integration, retrieval, agent memory and human oversight.

Local AI infrastructure

IN USE

Local inference · Model APIs · Workflow automation

Open-source language models served on local hardware and combined with external model APIs in everyday automation workflows — an architecture driven by privacy, performance and cost rather than defaults.

EXPERIMENTS

Trading automation · game development · art and visual design with generative AI

PROFESSIONAL EXPERIENCE

Associate Director / Team Lead, Lab IT

Dec 2021 – Mar 2026

BioNTech SE · Mainz

- Led a team of System Engineers delivering software and data analysis solutions for scientific R&D.
- Defined the system concept and served as System Owner for a biological assay data analysis platform operated across global laboratories, with responsibility for architecture, implementation and lifecycle management.
- Established the platform as a data provider for a closed molecule optimization loop — assay analysis results feeding directly back into personalized cancer therapy development.
- Translated scientific and business requirements into technical solutions, coordinating researchers, IT teams and external vendors in a regulated environment.

Senior Implementation Analyst

Sep 2018 – Dec 2021

Thermo Fisher Scientific · Dreieich

- Consulted on and implemented enterprise Laboratory Information Management Systems (LIMS).
- Analyzed laboratory workflows and translated business requirements into technical solutions.
- Delivered system configuration, validation and user training.

Automation Specialist

Dec 2015 – Sep 2018

Roche · Penzberg

- Designed and implemented software solutions supporting pharmaceutical production.
- Developed database logic and reporting services.

IT Consultant

Jun 2015 – Nov 2015

FERCHAU Engineering · Munich

- Contributed to projects automating pharmaceutical documentation.

EARLIER EXPERIENCE

Sep 2014 – Mar 2015 **Master Thesis Student** · Roche, Penzberg

Jan 2014 – Aug 2014 **Database Administrator** · University Medical Center Mainz

Sep 2013 – Jan 2014 **Research Assistant** · Johannes Gutenberg University Mainz

CORE COMPETENCIES

ARCHITECTURE & DELIVERY

System & Solution Architecture · System Ownership · Team & Technical Leadership · Stakeholder Management · Lifecycle Management

DATA & SCIENTIFIC SYSTEMS

Python · SQL · Databricks · Laboratory Informatics / LIMS · Scientific Data Management · Bioinformatics

AI & AUTOMATION

Agentic Workflows · LLM Integration · Retrieval & Agent Memory · Local AI Infrastructure · Workflow Automation

GOVERNANCE

Regulated Pharmaceutical Environments · System Validation · Data Governance

EDUCATION & LANGUAGES

M.Sc. Applied Bioinformatics
Johannes Gutenberg University Mainz
2012 – 2015

B.Sc. Biology
Johannes Gutenberg University Mainz
2009 – 2012

German — Native

English — Professional

Portuguese — Intermediate

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